MARK User Manual

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Windows x64 English Edition

MARK v1.0.0

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1 Introduction

MARK (Metadata-Guided Automation of Resource Workflows) searches, filters and downloads GNSS observations and precise products. It searches local file records for specified UTC dates. Select stations or products by sampling interval, equipment, satellite systems, date coverage and spatial conditions, then add the required data to a download plan. MARK supports data preparation for precise positioning, orbit determination, deformation monitoring, and ionospheric and tropospheric research.

This manual covers MARK v1.0.0 for Windows x64 with DuckDB knowledge bases. It explains installation, observation filters, spatial station selection, product presets, downloads and file management. The data cutoff is 2026-08-01 inclusive; coverage can end earlier or contain gaps. Screenshots illustrate the workflows. Counts and equipment records depend on the selected dates, sources and filters.

MARK provides two workbenches. The Observations workbench finds RINEX observation files by station and date. The Precise Products workbench finds orbit, clock, bias, ionosphere-map and other supporting files. Each workbench maintains its own save folder, download plan and task status. Queries and additions to the plan can continue while files are being transferred.

MARK prepares input data; it does not perform PPP, RTK or ionospheric parameter estimation. Before passing downloaded files to another application, check their format, constellation coverage, observation codes and compatibility with the processing model.

1.1 Main functions

(1) Search indexed observation records across global and regional sources, with source labels retained for each station.

(2) Filter stations by RINEX version, sampling interval, equipment valid on the requested dates, satellite systems, observation codes and completeness across the date range.

(3) Select study dates using Kp, Dst and F10.7, and restrict candidate stations by country or region, latitude and longitude, or map bounds.

(4) Select a target number of stations using even-spread or random selection, with optional distance and daily availability constraints. Inspect daily coverage reports.

(5) Search precise products by type, analysis center, product level and date-range completeness. Use built-in quick presets or save custom rules for several product types.

(6) Manage separate observation and product download plans. Pause, resume or cancel tasks, skip existing files, and try recorded alternative sources.

(7) Save files in year and day-of-year folders or in one flat folder. Optionally decompress observation files and convert Compact RINEX to ordinary RINEX.

1.2 Local queries and online downloads

The knowledge base stores file availability, filenames, station equipment, product attributes and download routes. It normally does not contain the original GNSS file contents. Queries for indexed dates use local records without repeatedly listing remote directories. When downloading starts, MARK contacts the data providers' servers to transfer files. A successful query therefore does not by itself establish that a download will succeed.

Local records are an archive snapshot taken for a release. Remote files may be moved or withdrawn, and access conditions may change. Missing local records can also reflect dates that have not yet been scanned.

Interpret available, missing, unverified and partially scanned source messages in the context of knowledge-base coverage.

2 Concepts and filter logic

2.1 Observation files and station days

Observation files contain GNSS measurements recorded by a station receiver. RINEX 2, 3 and 4 have different header and naming conventions. MARK groups RINEX 3 and 4 in one version-filter option. Providers may distribute compressed or Compact RINEX files, which must be prepared for the requirements of the subsequent processing application.

A station day is a station's record for one UTC date. MARK matches file availability, sampling interval, equipment and other conditions on the same date, then aggregates qualifying dates into station results. A station can become a candidate if at least one date satisfies the current filters. Set qualified date coverage to 100% when every effective date must qualify.

Site Logs and extended historical metadata describe station positions, receivers, antennas and observation capabilities. Equipment filters use records valid on the requested dates rather than simply applying the current model to the entire history. Missing metadata does not establish that an observation file is absent. When an include filter requires that metadata, an unknown record cannot automatically satisfy it.

2.2 UTC dates and day of year

All query dates use UTC, and both endpoints are included. Identical start and end dates define a one-day task. Day of year (DOY) is the date's ordinal position within the year. Save paths use three digits: for example, 2024-01-01 corresponds to 2024\001. A leap year has 366 days.

Advanced date filters can reduce the effective set of dates before station-range completeness is calculated. If a 30-day query retains only six dates that satisfy the space-weather conditions, completeness is calculated over those six dates. It does not describe file availability on all 30 original dates.

2.3 Two meanings of completeness

Table 2.1 Completeness measures

Measure	What is counted	Interpretation
Station or product range completeness	Available days in the requested date set	Stations must satisfy the current interval, equipment and other conditions on each counted date. Products are counted from their date records.
RINEX observation completeness	Actual versus planned observations within a file	An IGS/MGEX quality metric describing observation records. It differs from the number of dates with available files.
Daily availability constraint	Daily station counts in a fixed selected set	Each effective date must have at least the requested number or proportion of available selected stations. Different stations may be missing on different dates.

Single-station range completeness = qualifying dates ÷ effective query dates × 100%. If 27 of 30 dates qualify, completeness is 90%. If the receiver changes on day 10 and the new model does not satisfy the selected equipment filter, that day is not counted as available under that filter.

A file-date completeness of 100% does not guarantee all epochs, all required signals or sufficient processing quality in each daily file. When file contents are critical, check both the quality metrics and the actual RINEX files.

2.4 Combining filters and handling unknown records

Table 2.2 Rules for combining filters

Condition	Combination rule
Different filter categories	All enabled date, file, equipment and spatial conditions must be satisfied together.
Included satellite systems	AND: all selected systems are required on the corresponding date.
Included receiver or antenna models	OR within each equipment category: any selected model is sufficient.
Included observation codes	AND across selected codes, with each code interpreted within its satellite system.
Exclusions	A match to any enabled exclusion removes the corresponding record.
Enabled quality metrics	The relevant record must exist and meet the threshold. Unknown records do not pass automatically.

Unverified records are excluded from the confirmed available counts in station-set coverage reports. The table and map can also show different counts: a station without valid coordinates may appear in the table but cannot be plotted or used for station selection that depends on distance. The Runtime panel reports the counts and reasons.

3 System requirements and installation

3.1 System requirements

Table 3.1 Runtime requirements

Item	Requirement
Operating system	Windows 10 or Windows 11, x64.
Interface runtime	Microsoft Edge WebView2 Evergreen Runtime, x64.
Disk	About 10.6 GB installed. Setup calculates free space needed for the selected location, including temporary preparation files. Downloads and postprocessing need additional space.
Network	Indexed dates can be queried offline. Downloading original observations and precise products requires a connection.
Permissions	Readable application folder and writable download folder. Normal use does not require administrator privileges.
Other dependencies	The official package includes application runtime files and observation postprocessing tools. Python and a build environment are not required to run it.

If MARK reports a missing WebView2 runtime, install Microsoft's x64 Evergreen Runtime and restart MARK. Installing the Edge browser alone, or copying only the EXE, does not ensure that the required runtime and resources are present. Microsoft's download page is https://developer.microsoft.com/en-us/microsoft-edge/webview2/.

3.2 Downloading and installing MARK

(1) Open https://github.com/qiliu2025/MARK/releases and download MARK-v1.0.0-Windows-x64.exe. Code → Download ZIP provides the source repository; it does not include the complete runnable application and knowledge bases.

(2) Run the installer and choose an empty installation folder. Setup calculates the required free space for that location, including temporary preparation files. It displays elapsed time and estimated time remaining while preparing the knowledge bases.

(3) After installation, keep software and knowledge-base together inside the selected folder. The installation also includes documentation and source subfolders.

(4) Launch MARK from its shortcut or run software\MARK.exe. Each new installation starts in English and remembers subsequent language choices.

(5) Check a query and download for one station and one date before expanding the task.

The installer includes 64 DuckDB knowledge bases, runtime dependencies, observation postprocessing tools, the English PDF user manual and source/MARK-v1.0.0-source.zip. No Python installation or separate 7-Zip installation is required. The source archive includes MARK source, build scripts and required dependencies.

Table 3.2 Contents of the runtime package

Path relative to the MARK root	Purpose
software\MARK.exe	Desktop application entry point.
software\duckdb.dll	Database runtime; retain it alongside the EXE.
software\resources\	Interface, offline map, reference material and observation postprocessing tools.
software\curl-libressl.exe and supporting files	Network and runtime helpers; keep the supplied files together.
documentation\ and source\	English PDF user manual; source/MARK-v1.0.0-source.zip contains the corresponding application source and dependencies.
knowledge-base\	Official knowledge bases and supporting metadata; preserve the subfolder structure.

The installed application and knowledge bases occupy about 10.6 GB. Setup displays the space required at the chosen location; preparation also uses temporary space. Reserve additional space for downloads and expanded observations. To relocate MARK, move the whole installation folder with its subfolders.

3.3 Licensing and data use

Original MARK source is available solely for noncommercial scientific research under LICENSE. Researchers may inspect, modify and compile it for that purpose; commercial use requires prior written permission

from the MARK copyright holders. This is a source-available research license. The supplied knowledge bases have separate terms and may be used through the official client for noncommercial scientific research; modification or redistribution of the protectable database is not granted.

Third-party software and reference data retain their original licenses and rights. Original GNSS files downloaded from external networks and institutions remain subject to their providers' use, attribution and redistribution conditions. Consult LICENSE, KNOWLEDGE_BASE_LICENSE.md and THIRD_PARTY_NOTICES.md in the source repository for the full terms.

4 Main interface and first download

4.1 Workbenches and runtime messages

Use Observations and Precise Products at the upper right to switch workbenches. The observation page places filters on the left, the map on the right and the station table below. The product page places filters above the product table and details. The download bar belongs to the current workbench and shows the save folder, plan access, progress and task controls. Figure 4.1 shows the observation interface.

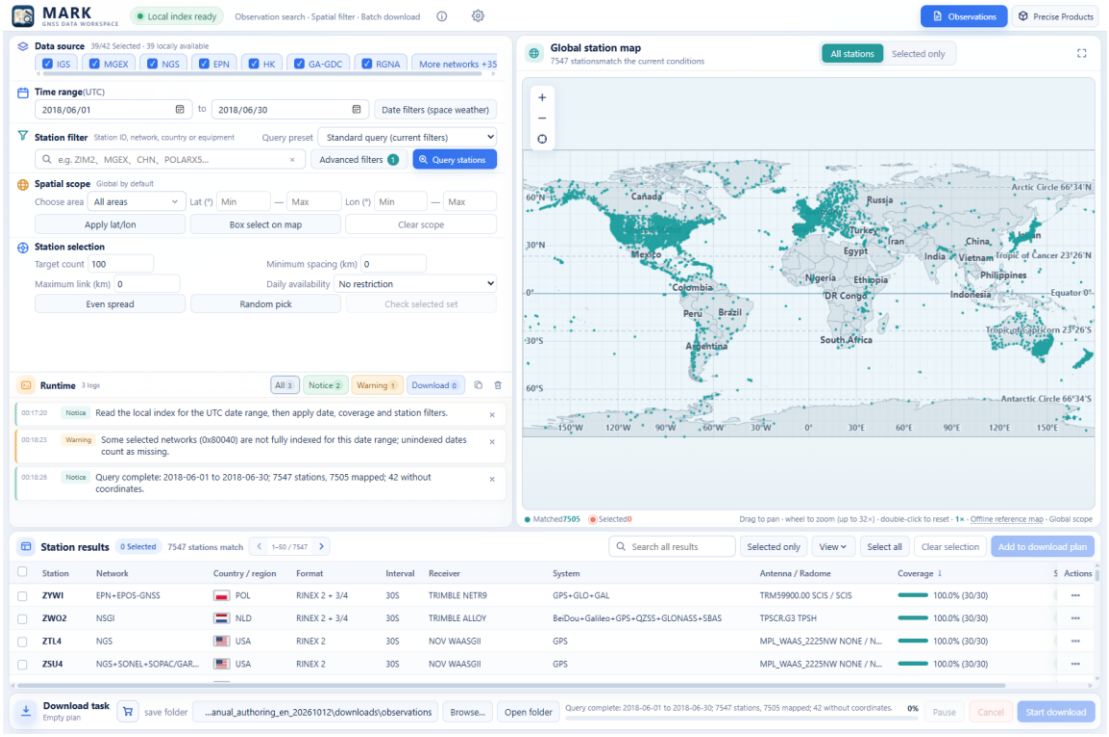


Figure 4.1 MARK Observations workbench

The Runtime panel separates notices, warnings and download messages. It shows query summaries, transfer failures and download-log paths. Completed or cancelled download items save file results, routes, errors and completion counts in the save folder’s logs subfolder. Copy retains the displayed messages; clearing Runtime does not delete the saved logs. Local index ready does not imply complete records for every source and date. About MARK shows the version, repository link and one knowledge-base data cutoff.

4.2 Language and download settings

Settings provides English and Chinese interface languages, Light and Dark appearance, and download-file options. Each new installation starts in English and remembers later language and appearance choices for that installation. Chapter 9 explains the file-saving options shown in Figure 4.2.

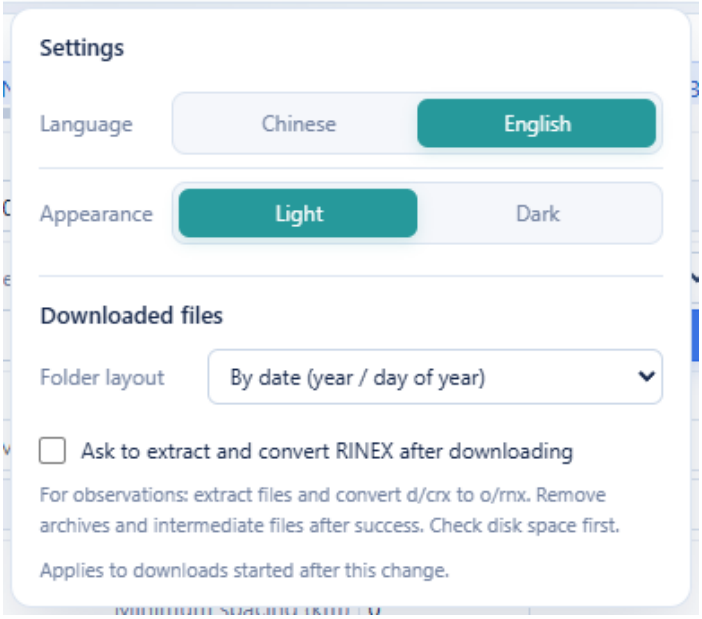


Figure 4.2 Language, appearance and download settings

4.3 Downloading one station for one day

This first-run example uses IGS station ZIM2 on 2024-01-01. Disable unrelated filters so that earlier conditions do not affect the result. Figure 4.3 shows the query settings.

(1) Select IGS on the observation page. If several networks are selected, open the complete source picker, clear the currently indexed networks with its select-all toggle, then select IGS alone.

(2) Set both dates to 2024-01-01. Open Advanced filter, click Restore defaults, select interval 30S and RINEX 3 / 4, then click Apply and refresh results. The default interval is 30S, not unrestricted.

(3) Enter ZIM2 in the station search and click Query stations. Check the network, format, interval and date-range completeness in the result.

(4) Select ZIM2 and click Add to download plan. Open the shopping-cart icon in the bottom bar to check the station ID and dates.

(5) Click Browse to select a writable save folder. Confirm that the plan item is checked, then click Start download.

(6) Inspect the successful, existing and failed counts and the download-log path in Runtime. Check the saved file and its format before using it in another application.

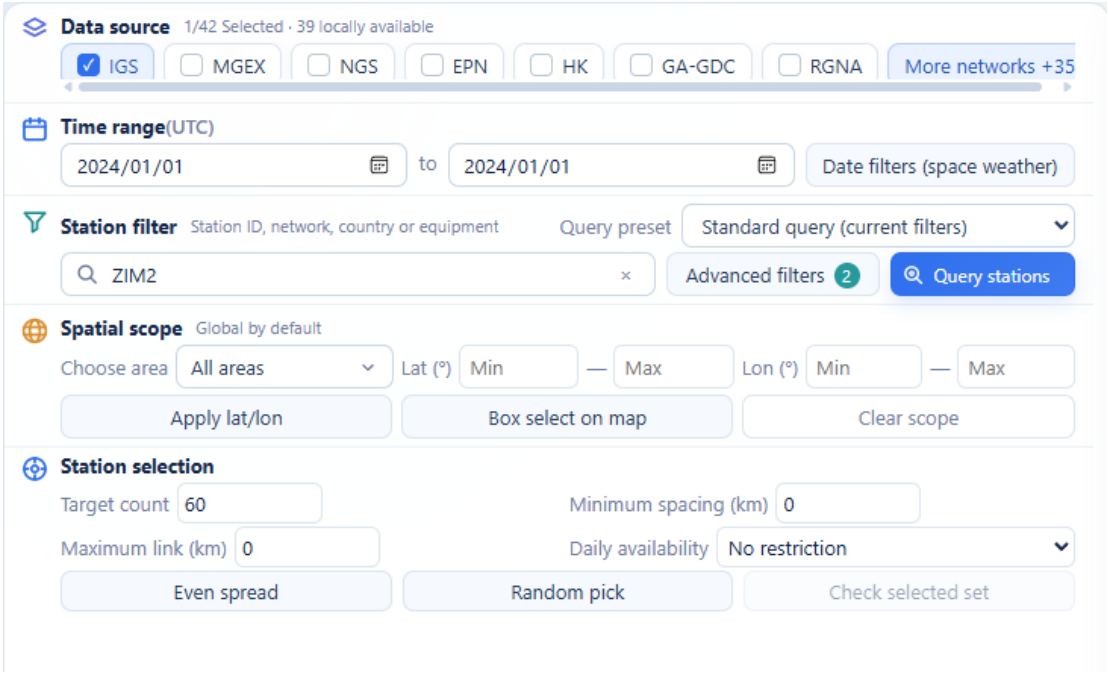


Figure 4.3 Query settings for ZIM2 on a single date

The verified task retrieved ZIM200CHE_R_20240010000_01D_30S_MO.crx.gz into <save folder>\2024\001\. Its header identifies Compact RINEX 3.0 with underlying observations in RINEX 3.02. Decompress and convert the .crx.gz file as required by the next application; see Section 9.2.

5 Observation searches and advanced filters

5.1 Selecting observation sources

The main page shows commonly used sources. Open the complete source picker to browse or search all options. On startup, MARK automatically selects sources with usable local knowledge bases. This release exposes 42 source options, of which 39 are available in the tested official package. These are interface options, not necessarily distinct physical networks. Figure 5.1 shows the picker.

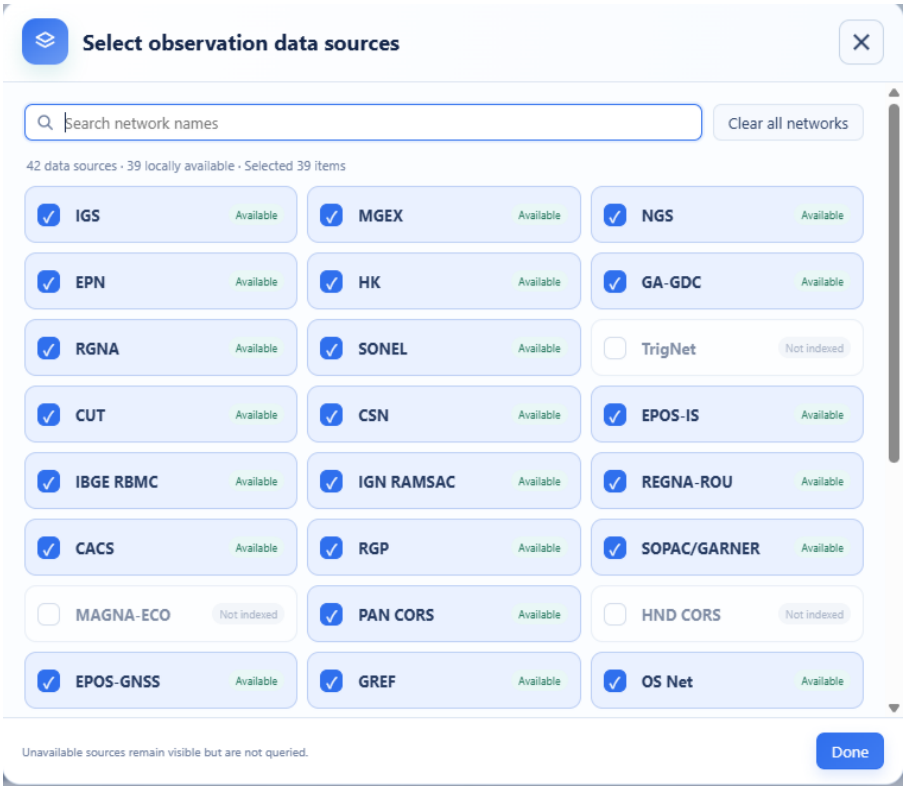


Figure 5.1 Source selection and local index status

Available means that the corresponding local index can provide records; check coverage for the requested dates separately. Unindexed sources remain visible, but selecting them does not build a knowledge base. When several sources identify the same physical station, MARK attempts to merge the records while retaining source labels. Identical four-character IDs can also refer to different entities, so the short ID alone is insufficient to identify duplicates.

5.2 Date filters and space weather

After setting the UTC dates, click Date filters (space weather). MARK first filters dates using the space-weather conditions, then queries observations and calculates completeness. If Kp, Dst and F10.7 conditions are enabled together, each retained date must satisfy all enabled conditions. Figure 5.2 shows the dialog.

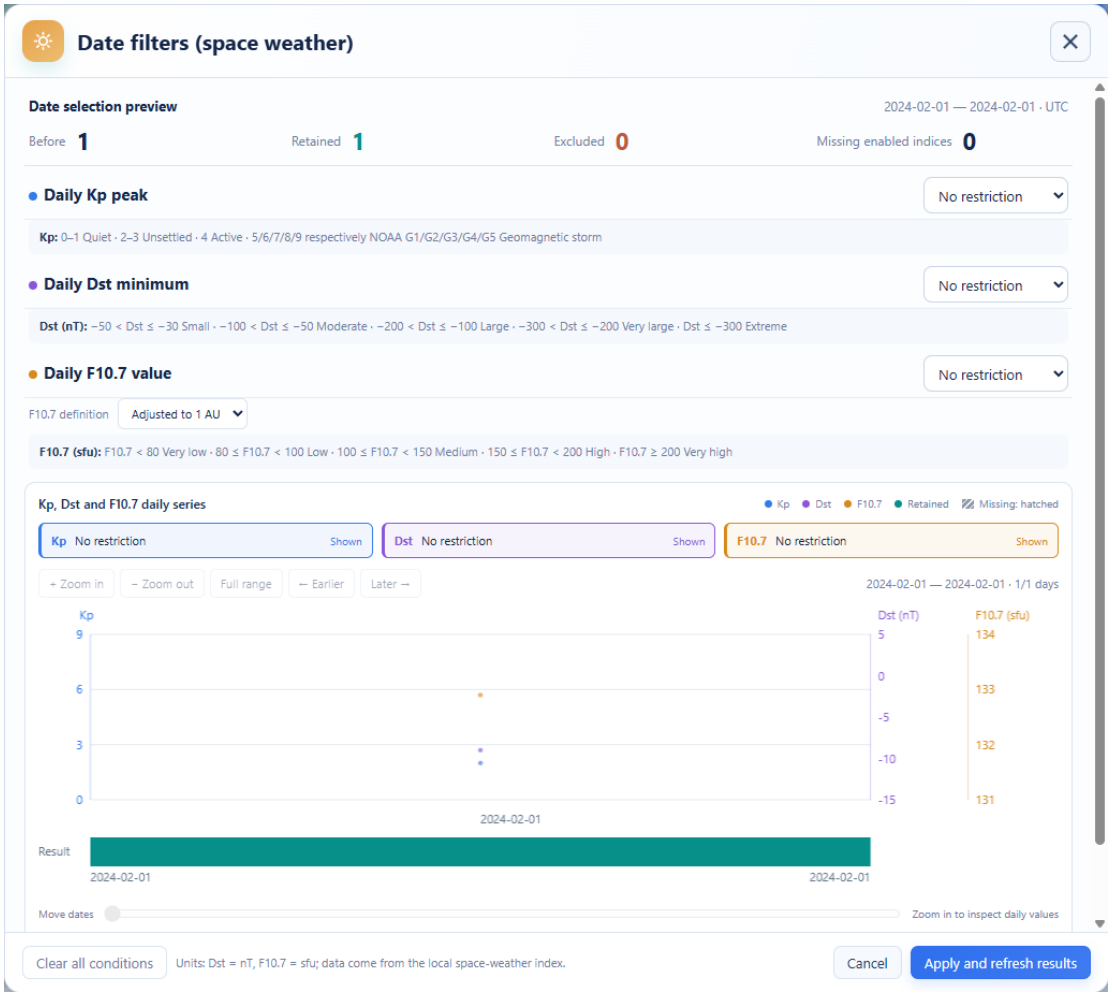


Figure 5.2 Date filters for Kp, Dst and F10.7

Table 5.1 Space-weather statistics and thresholds

Index	Daily statistic and unit	Example
Kp	Daily peak, 0 to 9	Keep > 4 retains dates whose daily peak is strictly greater than 4. The value 4 itself is excluded.
Dst	Daily minimum, nT	Keep ≤ −100 retains dates reaching −100 nT or a more negative value. More negative values generally indicate stronger disturbance.
F10.7	Daily solar radio flux at 10.7 cm, sfu	Filter by a range or a lower threshold. Interface groups provide selection guidance.

Exclude removes dates that meet the specified disturbance condition. A retained range uses the entered lower and upper bounds. Dates without a required index record cannot automatically satisfy the condition; inspect missing-data and exclusion notices. These indices help define a research sample and do not replace scientific analysis or an operational warning system.

5.3 Station completeness and observation format

Open Advanced filters. Under Availability and observation format, set qualified date coverage, minimum qualified days, interval and RINEX version. Coverage is the fraction of effective dates that pass all enabled file, equipment and quality conditions. Choose a minimum ratio, a minimum day count or No restriction. Applying the settings refreshes local results.

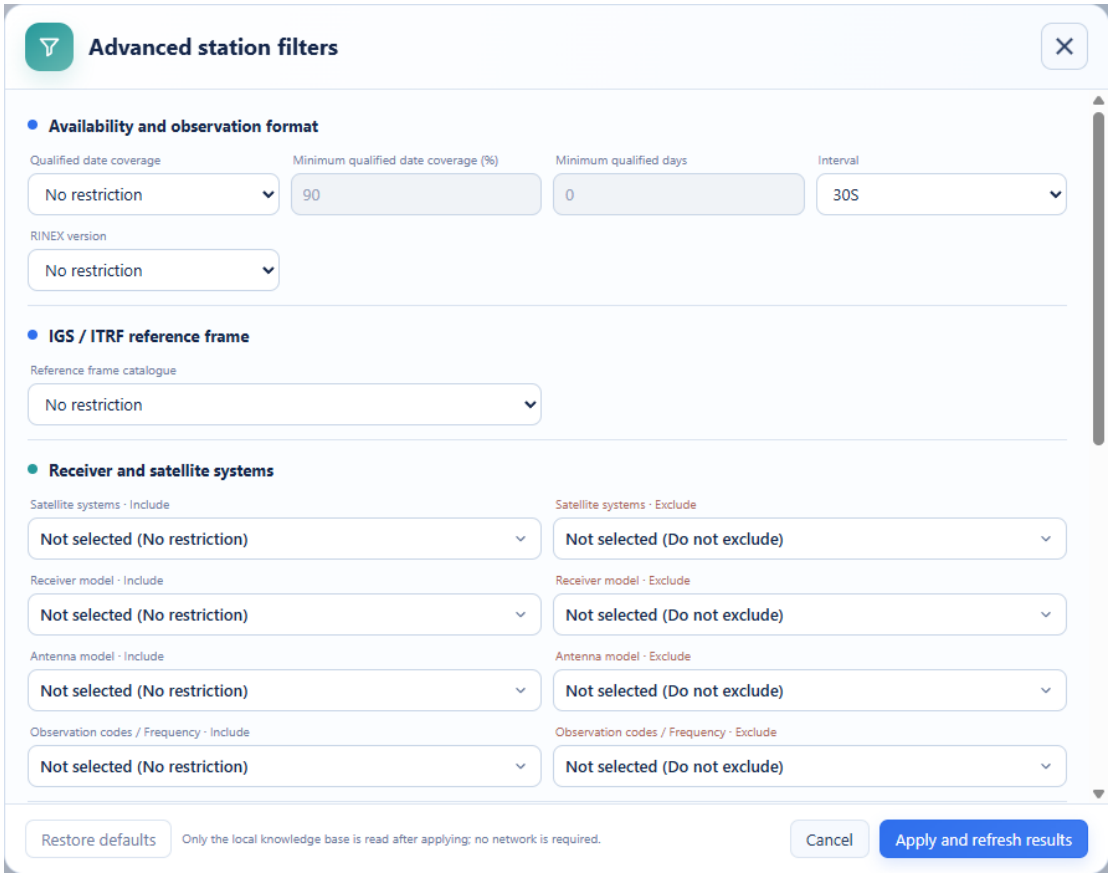


Figure 5.3 Advanced availability, format and equipment filters

The interval is the spacing between observations, for example 1S, 5S, 15S, 30S or 60S. RINEX version options are No restriction, RINEX 2 and RINEX 3 / 4. File organization and historical coverage of high-rate data vary between networks. Selecting 1S does not ensure complete daily files from every source.

5.4 Equipment models and observation signals

Use Include and Exclude controls to search and select receiver models, antennas or satellite systems. Apply and refresh results uses the selected conditions for the query. Figure 5.4 illustrates a receiver search.

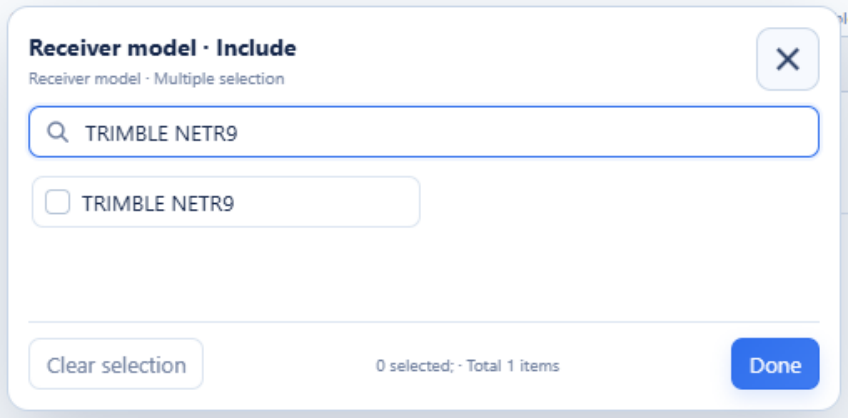


Figure 5.4 Searching for the TRIMBLE NETR9 receiver

System abbreviations include GPS, GLO, GAL, BDS, QZS, SBS and IRN. All selected include systems must be present, whereas any one of the selected include receiver or antenna models is sufficient. Equipment can change during the requested period. Use qualified date coverage to control how many dates must satisfy the equipment conditions.

Interpret observation codes together with their systems, such as GPS:C1C and BDS:C2I. Different systems can use the same code name. Code evidence varies across sources and historical dates; detailed signal records in this release mainly cover IGS/MGEX. Confirm that the chosen sources have the necessary records before enabling strict include filters.

5.5 Reference frames and GIM contributing stations

IGS / ITRF reference frame can automatically select the date-valid core-station catalogs for IGS14, IGb14, IGS20, IGb20 or IGc20, or use a specifically selected frame. This filters official catalog membership. It does not download a coordinate solution, transform observation coordinates or replace equipment-quality filters. Date ranges spanning several frames use the union of stations in overlapping catalogs; read the validity notices.

GIM station filters include or exclude known contributing stations by provider and product grade. Select GIM providers first, then their grades. The filter uses the union of station identifiers over the entire query range rather than matching contributors separately to each observation day. Products reporting counts without station identifiers are excluded from this filter. For independent evaluation, also check the GIM sources and their contributing-station information.

5.6 IGS and MGEX observation quality

The lower part of Advanced filters provides six daily quality criteria for IGS/MGEX stations with the required records. Metrics are evaluated separately for each UTC date, without averaging across dates. Values come from the local quality knowledge base, using the public IGS Network quality charts; MARK does not calculate these metrics from newly downloaded files. Each effective date must meet every enabled criterion.

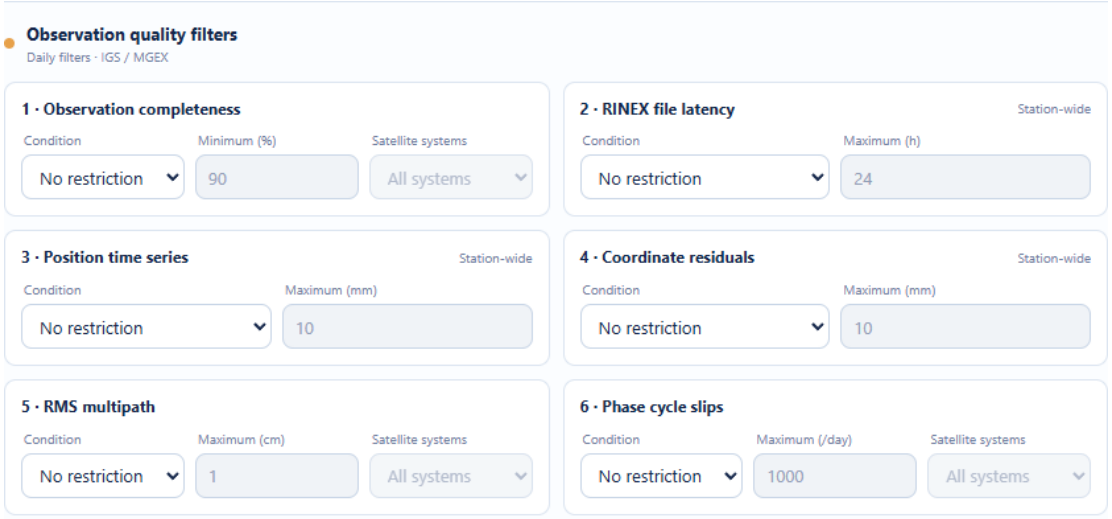


Figure 5.5 Observation quality filters

Table 5.2 Quality metrics and threshold directions

Metric	Definition and unit	Passing condition
Observation completeness	Daily actual / planned observations, %; All systems combines the totals, or select one constellation.	Observation completeness ≥ threshold.
RINEX latency	Daily file latency, h; station-wide.	Daily latency ≤ threshold.
Position time series	Daily mean absolute detrended east, north and up components, mm; station-wide.	Daily mean absolute displacement ≤ threshold.
Coordinate residuals	Daily mean absolute east, north and up residuals, mm; station-wide.	Daily mean absolute residual ≤ threshold.
RMS multipath	Daily mean of valid frequency-band RMS values, cm; select All systems or one constellation.	Daily mean ≤ threshold. Zero is treated as missing.
Phase cycle slips	Daily count for one constellation or sum for All systems, per day.	Daily total ≤ threshold.

Quality filtering reduces the qualified dates used for station coverage and downloading. A station can remain eligible when some dates fail unless the requested coverage requires every date. Strict combinations can reduce the result substantially. To investigate an empty result, disable quality filters and re-enable them one at a time.

Daily values and missing records. For observation completeness, MARK divides actual observations by planned observations for the selected constellation and multiplies by 100. All systems uses the combined actual and planned totals. A missing actual count or a non-positive planned total cannot establish completeness. RMS multipath is the mean of valid frequency-band RMS values for the selected constellation; All systems combines the valid values. A zero multipath value is an upstream unavailable-data placeholder and is excluded from the mean. Phase cycle slips use the daily count for the selected constellation; All systems sums the counts.

Station-wide metrics and units. Latency is measured in hours. Position time series uses the mean absolute value of the valid detrended east, north and up components on that day, in millimetres. Coordinate residuals use the mean absolute value of valid east, north and up residuals on that day, in millimetres. Multipath is measured in centimetres; observation numbers and cycle slips are counts. The daily means refer only to components or frequency bands within one date. Zero cycle slips, zero latency, zero displacement and zero residuals can be valid values.

Constellation selection. Observation completeness, RMS multipath and phase cycle slips each have an independent selector: All systems, GPS, GLONASS, Galileo, BeiDou or QZSS. QZSS corresponds to the upstream QZS label. These choices are independent of the satellite-system include and exclude filters for equipment capability. If a selected constellation lacks the required metric, that date does not qualify; another constellation cannot supply a replacement value. Latency, position time series and coordinate residuals are station-wide and have no constellation selector. Quality metrics cannot currently be filtered by observation code.

Qualified dates and coverage. MARK first matches file availability, RINEX version, sampling interval and date-valid station attributes, then intersects those candidate dates with every enabled quality condition. Only dates that satisfy all enabled conditions count as qualified. A metric that is not enabled has no effect when its record is absent. An enabled metric with no valid record makes the date unqualified, but an unknown metric does not establish that the file is scientifically poor.

Coverage denominator. Qualified date coverage = qualified dates / effective query dates × 100%. Dates removed by space-weather filters, reference-frame validity or other date restrictions are excluded from the denominator. Dates that fail quality thresholds or lack required quality records remain in the denominator. Missing quality records therefore cannot improve coverage by shrinking the denominator. A 100% setting requires every effective date to qualify; for 30 effective dates, a 90% setting requires at least 27 qualified dates.

Dates used by the results and downloads. Date calendars, qualified-day counts, coverage ratios, the dates used for even-spread station selection and the observation files selected for download all use the quality-filtered date set. The latest daily snapshot cannot stand in for the quality record on a different date. Saved presets retain their threshold values; each new query evaluates them using these daily rules.

Quality data sources. An example public station page is https://network.igs.org/ABMF00GLP. Its quality data interface is https://network.igs.org/network_qc?id=ABMF00GLP&type=<type>. The six chart identifiers are rnxobs (RINEX Observations), latency (RINEX Latency), timeseries (Position Time Series), residuals (Residuals), rms-multipath (RMS Multipath) and cycle-slips (Total Phase Cycle Slips). The IGS SLM project is available at https://github.com/International-GNSS-Service/SLM. These references identify the upstream sources; MARK applies the local daily statistics and units defined in Table 5.2.

5.7 Observation query presets

Query preset under Station filter applies a visible, editable group of conditions for common research tasks. Selecting a preset updates the controls; click Query stations to run the search. A preset does not download files. Review the selected stations before adding a download plan. Table 5.3 summarizes all six choices in this release.

Table 5.3 Observation query presets and default conditions

Preset	Main query conditions	Selection settings and action
Standard query	Current sources and advanced, date and spatial conditions.	No automatic selection. Switching to this preset does not restore all filter defaults.

Preset	Main query conditions	Selection settings and action
IGS/ITRF reference frame	IGS; 30S; RINEX 3/4; date-valid official core-station catalogs.	Restores global scope. Does not automatically select a fixed-size set.
GIM ionosphere stations	IGS + MGEX; global; 30S; GPS; keep GIM contributors from eight providers.	Target 250; automatic even spread after query; distance limits 0.
GIM evaluation	All locally available sources; global; 30S; RINEX 3/4; GPS; station completeness ≥90%; exclude GIM contributors.	Fixed 50 stations; minimum spacing 50 km; at least 35 confirmed available daily; automatic even spread.
Multi-GNSS POD	IGS + MGEX; global; 30S; RINEX 3/4; GPS, GLO, GAL, BDS; observation completeness ≥95%.	Target 60; automatic even spread; distance and daily constraints unrestricted by default.
Network RTK candidates	All locally available sources; 30S; RINEX 3/4; GPS + GLO.	Retains user scope and target count; minimum spacing 0, maximum link 75 km; run Even spread manually.

(1) Standard query (current filters)

Use this option for station IDs, equipment searches or custom conditions. It retains the current networks, interval, equipment and geographic scope rather than imposing one research template. Switching to Standard query clears automatic reference-frame selection and preset daily constraints, but existing equipment, quality and spatial filters can remain. For an unrestricted search, also restore advanced defaults and clear the station search and spatial scope.

(2) IGS/ITRF reference frame (date-aware)

Use this preset to prepare observations from official reference-frame core stations. It selects IGS, 30S and RINEX 3/4, restores global scope, and chooses IGS14, IGb14, IGS20, IGb20 or IGc20 core catalogs for the query dates. A range spanning a frame transition uses the union of stations in overlapping catalogs. Station date-range completeness and all six quality criteria are unrestricted by default. Add 100% station completeness if each core station must have files throughout the period.

The preset filters official core-station membership. It does not download a coordinate solution or transform coordinates. Select the required stations after querying; it does not automatically impose a 50-, 60- or 250-station set.

(3) GIM ionosphere stations

Use this preset to find observations associated with GIM modelling. It selects IGS and MGEX, global scope, 30S and GPS, with no restrictions on RINEX version, equipment, observation codes or station date-range completeness. It keeps contributors recorded for IGS, CODE, ESA, JPL, UPC, CAS, EMR and WHU, using all product grades by default. Contributor IDs are combined across the entire query period. GIM records containing counts without station identifiers do not contribute to the filter.

The target is 250 stations, with both distance limits set to 0. An even-spread selection runs automatically after querying; candidate counts and coordinate availability can limit the result. Daily availability in the current selection controls is unrestricted by default. If the study requires at least 245 stations daily, explicitly set that threshold under Daily availability, rerun Even spread and check the set. Do not infer that the daily threshold has passed solely from the preset name or a notice.

(4) GIM evaluation (even spread and daily availability)

Use this preset to prepare evaluation observations separated, as far as the recorded contributor data permit, from GIM modelling samples. It selects all locally available sources, global scope, GPS, 30S and RINEX 3/4, and requires at least 90% station date-range completeness. It excludes stations used by the eight providers on any date in the period. All grades are included in that exclusion by default; adjust providers and grades in Advanced filters when needed.

After the query, MARK automatically selects one fixed set of 50 stations with at least 50 km spacing and at least 35 confirmed available selected stations on every effective date. Dates below threshold remain in the report rather than being removed to make the constraint pass. Missing contributor records can make exclusion incomplete. Check independence against the actual GIM products and station lists used in the study.

(5) Multi-GNSS POD (60 global stations)

Use this preset to prepare candidate observations for multi-GNSS precise orbit determination. It selects IGS and MGEX, global scope, 30S and RINEX 3/4, requires GPS, GLONASS, Galileo and BDS capabilities together, and enables the IGS/MGEX RINEX observation-completeness criterion at 95% or above. This metric measures actual versus planned observations within files. It does not mean that a station has files on 95% of query dates. Station date-range completeness is unrestricted by default, and stations without the required quality records do not pass.

An even-spread selection with a target of 60 stations runs after querying. Distance and daily availability constraints are unrestricted by default; add them if a stable daily network or full-month station set is needed. The downloaded files retain their original 30S interval. A POD processor can decimate them to 300S when required; this preset does not rewrite observations.

(6) Network RTK candidates (regional)

Use this preset to screen historical observations for a regional candidate network. It selects all locally available sources, 30S, RINEX 3/4 and GPS + GLONASS. All six IGS/MGEX quality criteria are disabled so that regional stations are not rejected simply for lacking those global quality records. Existing country, coordinate or map scope and target count are retained. Minimum spacing is 0 and Maximum link is 75 km.

Define the service area and target count, click Query stations, then run Even spread manually. The 75 km limit requires a connected graph using links no longer than that limit, rather than limiting every station pair to 75 km. Availability of historical daily files does not establish real-time uptime, NTRIP/RTCM service, latency or site quality. Verify those properties separately for deployment.

Figure 5.6 illustrates the GIM evaluation selection controls: one fixed set of 50 stations, minimum spacing 50 km and at least 35 confirmed available stations on every effective date. Candidate and selected counts depend on the date range and available records.

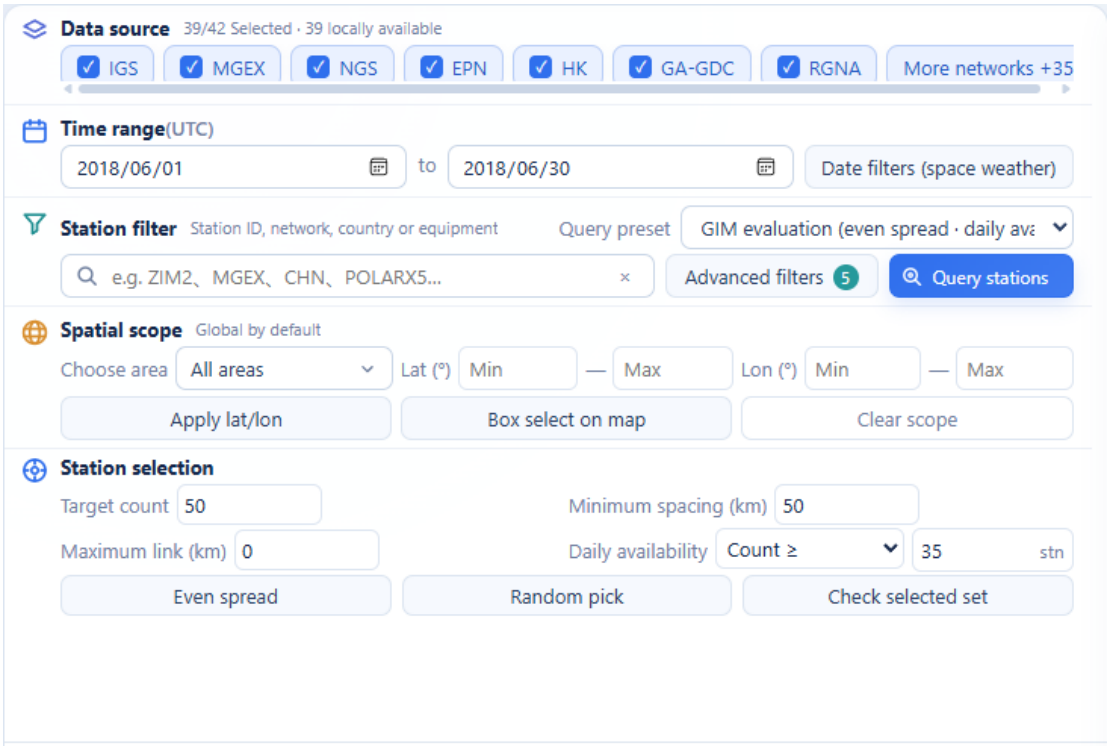


Figure 5.6 Default selection settings for GIM evaluation

Task presets overwrite several controls and normally clear equipment includes and exclusions, observation codes and space-weather conditions. Review the station search, advanced filters, scope, target and daily constraints after applying a preset. Query and check the set again after changing dates or conditions. GIM stations, GIM evaluation and POD run automatic even-spread selection; Standard query, reference-frame and Network RTK searches require station selection afterward.

6 Spatial scope and station selection

6.1 Countries and geographic bounds

Open the country/region picker, search or select the required regions, then click Apply country / region scope. Leaving the country selection empty uses the global scope. You can also enter latitude and longitude limits and click Apply lat/lon, or choose Box select and drag a rectangle on the map. Enabled country and coordinate filters apply together. Figure 6.1 shows the country picker.

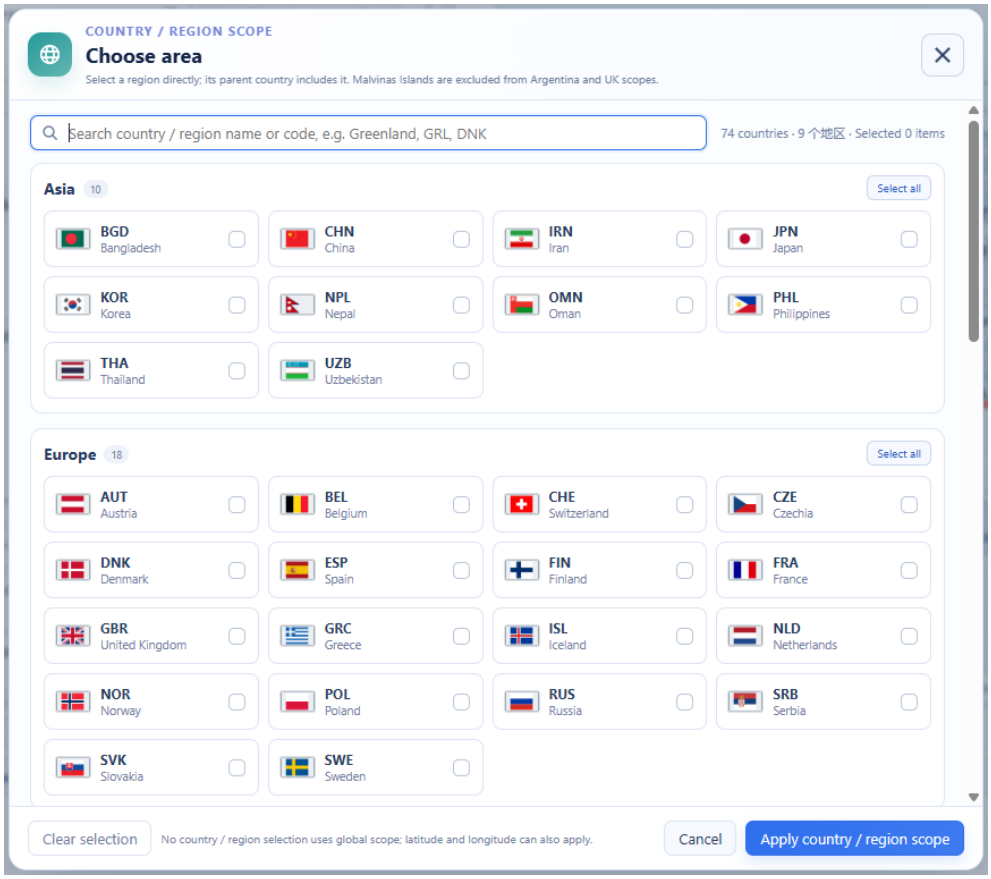


Figure 6.1 Selecting countries and regions

Latitude ranges from −90° to 90° and longitude from −180° to 180°. A longitude lower bound greater than the upper bound crosses the date line: 170° to −170°, for example, defines a narrow band across the Pacific. Clear scope restores global spatial bounds. After changing the scope, check the current selection and select stations again if necessary.

The map supports dragging to pan, wheel zoom, double-click reset and fullscreen display. Teal points are matching candidates and orange points are selected stations. Selected only changes the display. Click a map point or a table row to open station details. Records without coordinates cannot be plotted.

6.2 Even spread and random selection

Under station selection, enter the Target count and click Even spread or Random pick. Even spread progressively builds a dispersed set using spherical distances, suitable for global or regional sampling. Random pick supports random sampling and can produce a different set on each run. Distance and daily availability constraints affect which selections are feasible.

Table 6.1 Automatic station-selection parameters

Parameter	Meaning
Target count	Requested size of one fixed station set; a new set is not selected for each date.
Minimum spacing (km)	Minimum distance between any two selected stations. 0 disables the constraint.
Maximum link (km)	Selected stations must form a connected graph using links no longer than this limit. 0 disables the constraint.
Daily availability	Unrestricted, daily count ≥ or daily ratio ≥. Applies to confirmed availability in the selected set on each date.
Check selected set	Check whether the current manual or automatic selection satisfies the count, distance and daily constraints.

Maximum link (km) constrains network connectivity; it does not require every pair of stations to be closer than that distance. For global sampling it is usually set to 0. Regional Network RTK candidates can use a value appropriate to the operating area. MARK reports when the target count, minimum spacing or daily threshold cannot be met. Adjust the constraints or widen the candidate scope.

6.3 Checking and exporting daily coverage

Click Check selected set to inspect the set size, minimum daily available count, dates below threshold, actual minimum spacing and daily curve. Unverified stations are excluded from confirmed available counts. Dates below threshold are retained in the study date set. Figure 6.2 shows the check for a fixed set of 60 stations over a full month.

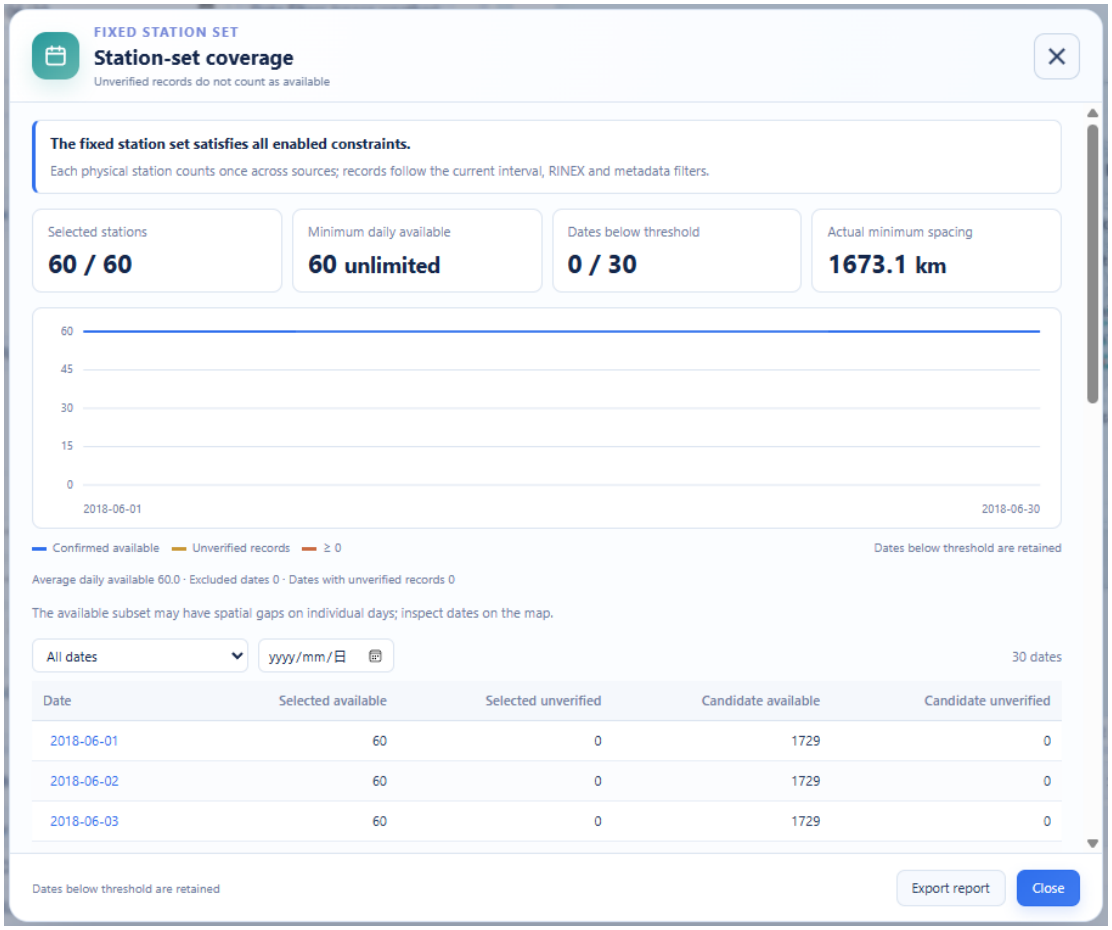


Figure 6.2 Daily coverage of a fixed station set

Click a date in the report to inspect its spatial distribution. An aggregate count alone does not establish even coverage on every day. Export report saves a CSV with station IDs, dates, and available and unverified selected counts. Changes to filters or station selection can invalidate the report; run the check again.

6.4 Station table and details

The result table supports searching within current results, showing selected stations only and sorting by range completeness. View controls columns and display density. Use checkboxes to select stations. Row clicks and action menus provide details, map location, selection changes and copying. Select all applies to the current filtered list. Verify the selected count and download plan before downloading.

Station details show identifiers, coordinates, sources and equipment. Four-character IDs and long identifiers serve different purposes; ZIM2, for example, has the long identifier ZIM200CHE. A dash (—) in an equipment field provides no evidence of the model or supported signals. Figure 6.3 shows actual station details.

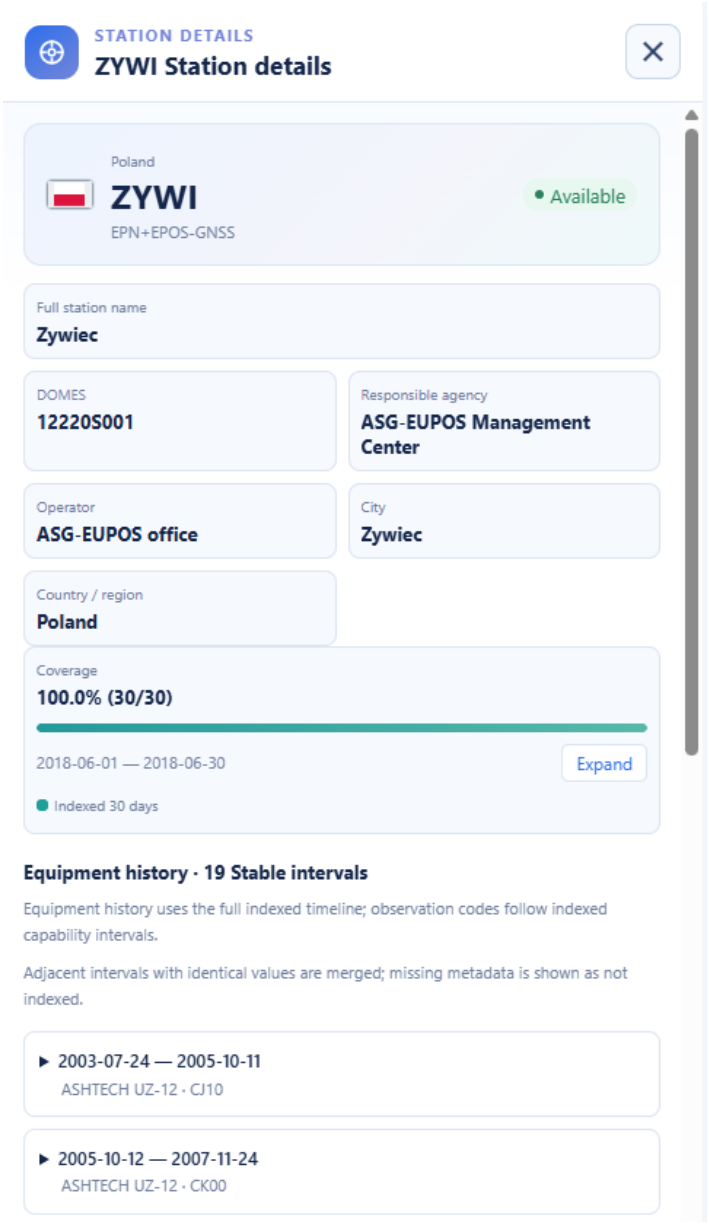


Figure 6.3 Station sources, completeness and equipment history

7 Precise product searches and presets

7.1 Product types and basic filters

Switch to Precise Products, set UTC dates, Product type, Product level, Analysis center and range completeness, then click Query products. The main filter selects one product type at a time or All Products. Use the search box to further filter product IDs. Figure 7.1 shows the workbench.

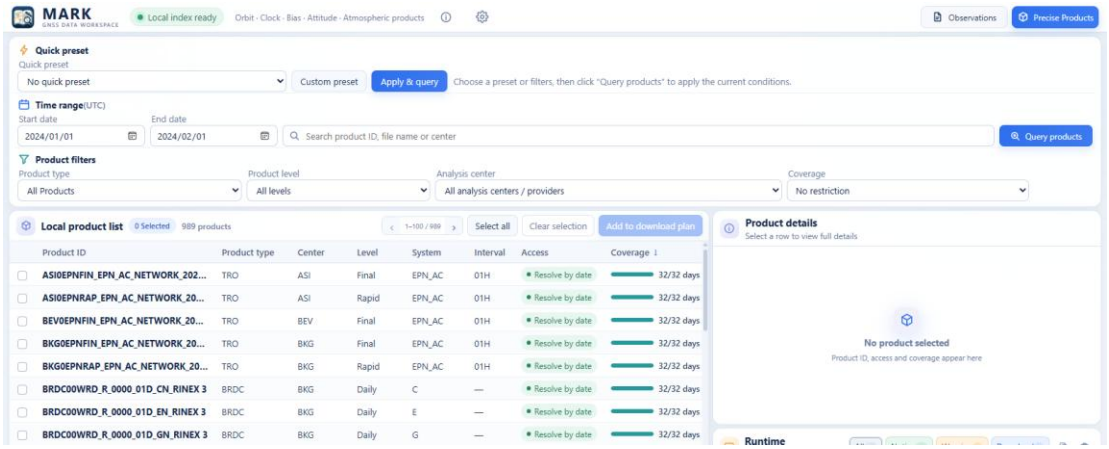


Figure 7.1 Precise Products workbench

Table 7.1 Product types and common uses

Type	Use	Common format or content
SP3	Precise satellite orbits	SP3.
CLK	Precise satellite clocks	RINEX clock.
ERP/EOP	Earth rotation and orientation parameters	Polar motion, UT1 and related parameters.
DCB	Differential code biases	DCB, Bias-SINEX and related formats.
OSB	Observable-specific biases	OSB, BIA, Bias-SINEX.
UPD	Uncalibrated phase delays	Wide-lane, narrow-lane and related bias products.
GIM	Global ionospheric maps	IONEX, INX, IOX and related formats.
TRO	Tropospheric parameters	ZPD, TRO-SINEX and related formats.
SINEX	Coordinate and solution products	SNX, SSC and related formats.
ANTEX	Antenna phase-center models	ATX.
ATT	Satellite attitude	ORBEX, ATT.OBX.
BRDC	Broadcast ephemerides	RINEX navigation files.
VMF	Tropospheric mapping-function grids	VMF1, VMF3 and related epoch grids.

Product level describes properties such as final, rapid, ultra-rapid, predicted or static publication; it does not describe download speed. The analysis center generates the product, whereas the download source supplies the archived file or mirror. These can differ. For example, indexed CODE products may be downloaded through CDDIS, IGN or WHU routes.

The main filter has no separate constellation or sampling-interval dropdown. Inspect these attributes in the product table and details. Custom presets support interval rules for applicable product types. Interpret VMF series such as OP and FC according to their own definitions rather than equating them directly with orbit levels such as Final or Rapid.

7.2 Product-series completeness and details

A table row generally represents a product series that can be resolved by date, rather than an already downloaded file. Fixed dates, 0000 and time-span fields in a series ID do not replace the current UTC query range. Actual filenames are resolved for each date when the download list is built. Range completeness describes the requested date records; 32/32 days, for example, is 100%.

Select a row to inspect its center, level, systems, interval, date coverage and download-source notes. Before combining SP3, CLK and bias products in one processing setup, check their product lines, reference frames, signals and bias conventions. A current ANTEX alias or similar filename alone does not establish compatibility.

CLK details show antenna-model declarations recorded in the knowledge base. If a declaration is found but its historical file version cannot be identified, the interface shows Model declaration; version unconfirmed. In this example, the GFZ CLK record for 2024-01-01 declares IGS20_2290. This does not establish that any current igs20.atx alias is that historical version. Figure 7.2 shows the declarations.

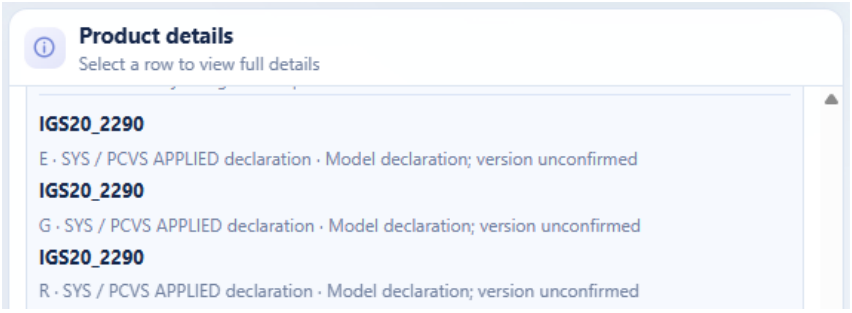


Figure 7.2 ANTEX model declarations in CLK headers

7.3 Built-in quick presets

Set the UTC start and end dates, choose a Quick preset and click Apply & query. Changing the dropdown alone does not build a bundle. Query products runs the ordinary filters and exits the active built-in preset. A quick preset uses one SP3 family as its anchor, matches the required members for each date, and reports complete-bundle coverage and missing products and intervals. Table 7.2 summarizes all six plans; the detailed rules follow.

Table 7.2 Product bundles and rules for built-in quick presets

Preset	Product bundle and main rules
PRIDE PPP-AR	Fixed WUM0MGXRAP rapid multi-GNSS products and 30S CLK; OSB, ERP and ATT; CLK-matched ANTEX; also checks SINEX, GIM, VMF and BRDC members.
IGS Final PPP	IGS Final OPS SP3 + 30S CLK + ERP/EOP + IGS GLOBAL ANTEX for standard float PPP preparation.
UPD PPP-AR	Matching producer and product family: SP3 + CLK + UPD + ERP + ANTEX; compares Final/Rapid candidates.
OSB PPP-AR	Same producer and product line: SP3 + 30S CLK + OSB + ERP + ANTEX; compares Final/Rapid candidates.

Preset	Product bundle and main rules
PPP-RTK	SP3 + 30S CLK + OSB + ERP + ANTEX base files; regional atmospheric corrections and real-time services are separate.
Automatic PPP	Compares SP3 + CLK + ERP + ANTEX bundles; prioritizes date coverage, then compatibility records, Final/Rapid and CLK interval.

(1) PRIDE PPP-AR

This plan applies explicit file-selection rules for data preparation with PRIDE PPP-AR. Core products are fixed to Wuhan University WUM0MGXRAP, the rapid MGX multi-GNSS line, with 30S CLK. SP3, CLK, OSB/BIA, ERP and ATT/ORBEX retain the specified producer and family; a missing member is not replaced by another producer. OSB uses the line's all-frequency product. A bias file for one frequency combination is not an equivalent substitute. Table 7.3 lists the core and supporting members checked together.

Table 7.3 Product-selection rules for the PRIDE PPP-AR preset

Member	Selection rule	Purpose or check
SP3 and CLK	WUM0MGXRAP; MGX; Rapid; CLK 30S.	Keep orbit and clock in one product family.
OSB/BIA	All-frequency OSB from the WUM rapid line.	Observation biases for ambiguity fixing; check signals and bias conventions.
ERP	WUM0MGXRAP rapid ERP.	Earth rotation parameters for the rapid family.
ATT/ORBEX	Attitude products matching the WUM rapid family.	Satellite attitude support.
ANTEX	Match declarations in the selected CLK for each date.	Prepare the calibration model used by the clocks; see below.
SINEX CRD and SOL	IGS0OPSSNX; daily; Final.	Coordinate and solution support; both classes are checked.
GIM	COD0OPSFIN final GIM.	Ionosphere support.
VMF1 and VMF3	OP; VMF1 2.5×2°, VMF3 1×1°; 00, 06, 12 and 18 UTC daily.	Tropospheric mapping grids; all eight model/epoch members are checked.
BRDC	IGS BRDC00IGS; MN; daily RINEX 3.	Mixed-system broadcast ephemerides.

In this release, the supporting members in Table 7.3 also count toward Complete bundle. The check includes more than orbit, clock and bias files. SINEX has CRD and SOL members, and each VMF model has four daily epochs, so the number of checked members exceeds the number of product groups in the table. A missing VMF epoch or other supporting member can make the bundle incomplete even when SP3, CLK and OSB exist. Inspect the list and download available files; the auxiliary products actually needed for processing depend on the PRIDE configuration and study.

ANTEX selection starts from the chosen CLK. For each query date, MARK reads the clock family's recorded header declarations and searches the local ANTEX catalog for the named file or historical model revision, such as IGS20_2290. Declarations can differ by system or change within the date range, yielding several ANTEX files; identical files are combined into one displayed entry. The plan does not simply choose the current igs20.atx from the observation date or apply the WUM producer filter to antenna calibrations.

If a CLK declares only a base model such as IGS20 without a historical revision, MARK can use the same model's official current file where the records permit, with a version-unconfirmed or compatibility notice. A current alias can change and is not proof of the file originally used to generate historical clocks. Missing CLK header records, declarations or a matching ANTEX with download routes produce an ANTEX gap for the affected date. MARK does not invent a revision.

First confirm the WUM rapid CLK returned by the plan. Open its details and inspect ANTEX referenced by CLK, then compare the declarations, systems, dates and matched files. An ANTEX row's range completeness counts dates on which the associated CLK references it; it does not describe the ANTEX file's internal validity period. Retain downloaded CLK and ANTEX filenames and hashes, and check models and signals against the PRIDE version and configuration. Resolve unconfirmed historical versions before formal processing.

Figure 7.3 illustrates the selected WUM0MGXRAP 30S CLK for 2024-01-01, with IGS20_2290.ATX declarations for C, E, G and R. Inspect the actual matched ANTEX entries and complete-bundle coverage returned for the requested dates.

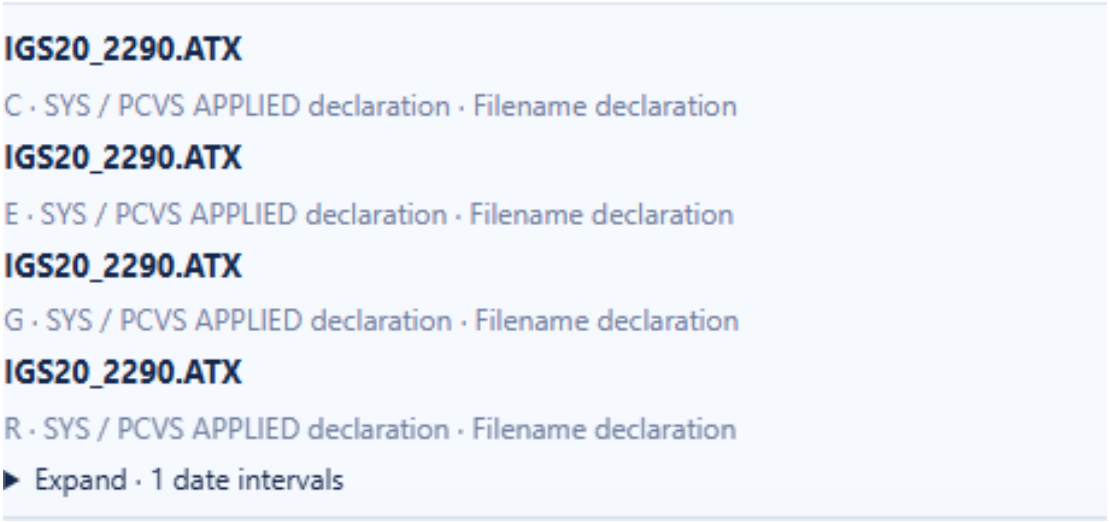


Figure 7.3 Historical ANTEX declarations in the CLK selected for PRIDE PPP-AR

(2) IGS Final PPP

This preset prepares IGS Final OPS SP3, 30S CLK, ERP/EOP and IGS GLOBAL ANTEX for standard float PPP. SP3 and CLK use the specified IGS family. ERP/EOP may come from an allowed compatible global Earth-orientation stream, such as CODE ERP; every auxiliary file need not be produced directly by IGS. ANTEX is selected using date-effective frame and model information. This preset does not run PRIDE's per-date CLK-declaration replacement step.

OSB and UPD for ambiguity fixing are not included, nor are ATT, GIM or VMF added by default. Query them separately when required. The verified 2024-01-01 example returned IGS SP3, IGS CLK, IGS20 ANTEX and matching CODE ERP, with complete coverage for 1/1 days.

(3) UPD PPP-AR

This preset compares SP3, CLK, the corresponding producer's UPD, ERP and ANTEX, using available family, frame, system and bias-convention records. Some UPD entries are local inventory records without an online download route. Such entries cannot be downloaded through MARK and must not be treated as online availability. Inspect the access status and returned clock interval.

UPD combinations such as WL, NL or EWL describe the bias solution. File existence alone does not make them compatible with arbitrary clocks. Confirm the UPD convention and observation combination supported by the downstream application. OSB and UPD are different representations and are not interchangeable bundle members.

(4) OSB PPP-AR

This preset selects SP3, 30S CLK, OSB, ERP and ANTEX from matching producers and product lines, comparing complete coverage among Final/Rapid candidates. Core producer, series, systems and known frequency or bias conventions must match. A bias product can cover a subset of the orbit systems but cannot introduce a system absent from the orbit.

Strict matching reports missing critical compatibility records. Date completeness of 100% does not replace a bias-convention check. Confirm the OSB representation, signals and reference model supported by the PPP-AR application; do not assemble orbit, clock and bias files across unrelated producers.

(5) PPP-RTK

This preset organizes the same base product classes as the OSB plan: SP3, 30S CLK, OSB, ERP and ANTEX, with family-compatibility checks. It prepares base files for relevant processing systems. Regional ionospheric or tropospheric corrections, SSR streams, NTRIP broadcasting and network estimation are supplied by the user's service or processing system.

Network RTK candidates in Observations selects stations; the PPP-RTK quick preset in Precise Products selects supporting products. These are separate preparation steps. Applying one does not configure the other.

(6) Automatic PPP

MARK compares SP3, CLK, ERP and ANTEX bundles from identifiable producers and product lines. It first prefers a compatible family covering more query dates. For equal coverage, it prefers more complete critical compatibility records, then Final over Rapid, and then shorter CLK intervals. It does not fill missing dates by combining unrelated orbit and clock producers.

A Rapid family covering the full period can therefore beat a Final family covering only part of it. Automatic PPP does not force Final or select the shortest CLK independently of the winning family. Check the returned list against the processing application. This plan organizes four classes for float PPP and does not automatically add OSB, UPD or regional atmospheric corrections.

Complete-bundle days count dates with every member required by the plan. Each table row's completeness describes that product alone. Review missing-date intervals, compatibility notices and row coverage together. Ordinary filters and custom presets combine search conditions without the built-in plans' bundle-selection rules. Reapply and query after changing the dates or plan.

7.4 Creating and reusing custom presets

(1) Click Custom preset, select New preset and enter a recognizable name.

(2) For each rule, select the product type, level, center and applicable sampling interval. Click Add product to add another rule. The interval control is disabled for types to which it does not apply.

(3) Click Save preset. The saved preset appears in the Custom section of the Quick preset list.

(4) Set a new date range, select the preset and click Apply & query. Check which returned product series are required.

(5) Reopen Custom preset to edit existing rules or delete the current preset.

Custom rules reuse filter conditions. Results from different rules are combined, and one rule can match several product series. A rule is not limited to one file, and matching results do not establish that products from different centers can be processed together. Presets are stored locally in WebView2 user data without changing the knowledge base. Recreate them after changing computers or clearing that user data. Section 10.2 gives a complete example.

8 Download plans and task management

8.1 From selection to a download list

Select stations or products, then click Add to download plan. Each entry retains its identifier, dates, save folder and relevant query conditions. MARK resolves the corresponding files when the entry starts. Selecting rows, adding entries and transferring files are separate actions. Observation and precise-product plans remain independent when switching workbenches. A changed save folder applies to subsequently added entries.

Open the shopping-cart icon in the bottom bar to inspect the plan. Check station or product IDs, dates and pending status, then select the entries to execute. Select all applies to every entry in the current workbench plan, including entries beyond the visible portion. Start download and Remove in this dialog apply to checked entries only. Removing a plan entry does not delete saved files. Figure 8.1 shows a completed station entry.

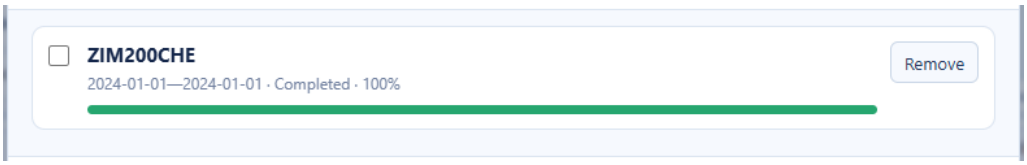


Figure 8.1 Station ID, dates and task status in a download plan

8.2 Parallel transfers and task controls

Several station or product entries can run concurrently, including different stations on the same date. Files within a multi-day entry can also transfer concurrently. Both workbenches share a global limit of 12 transfers; individual servers have lower adaptive limits. Running plan items can therefore exceed the number of actual network transfers. The download bar shows the active item count and the most recent file message; the plan shows each item’s status and progress.

Pause suspends downloads in the current workbench. The bottom-bar Cancel stops that workbench's whole run; completed files are retained. Checkboxes select entries for a subsequent start and do not stop a transfer already running. MARK does not restore plans or running jobs after closing; query and add them again after restarting. Existing files are checked before transfer. Pause and resume do not guarantee byte-level resume support on every server.

Inspect the successful, existing and failed counts and the saved download logs. Each entry records filenames, source routes, results and errors under the save folder’s logs subfolder; Runtime shows its log path on completion or cancellation. A full progress bar does not guarantee that every file succeeded. Resolve the reported error and rerun the affected entry when needed.

8.3 Save folders and existing files

Each workbench has its own save folder. The default date layout places observations at <root>\YYYY\DOY\<filename> and most precise products at <root>\YYYY\DOY\<product category>\<filename>. Static files such as ANTEX follow their category or center-specific rules and do not necessarily have a separate daily folder.

Table 8.1 Common precise-product folder categories

Product	Category folder
SP3 / CLK	orbit-sp3 / clock-clk
ERP/EOP / DCB	erp-eop / dcb
OSB / UPD	osb / upd
GIM / TRO	ionex / troposphere
SINEX / ANTEX	sinex / antex
ATT / BRDC	attitude-orbex / broadcast-navigation
VMF	vmf-grid, with model, grid and solution subfolders.

MARK checks existing files before skipping a transfer, including basic content and format checks. An already converted ordinary RINEX file can satisfy an observation entry. These checks do not replace scientific validation of epochs, signals or processing quality. Preserve and move aside a confirmed corrupt file before downloading it again.

8.4 Alternative sources and error diagnosis

A logical file can have several recorded sources. MARK makes limited retries and uses supported alternative routes when available. Japan GEONET observations use the supported anonymous Terras HTTPS workflow. SFTP sources require Windows OpenSSH Client and applicable source access. HTTP 404, FTP 550, login pages and permission changes are recorded in the task log and require checking the original provider.

Downloaded files retain their server filenames. The verified single-product example retrieved the following orbit file into <product folder>\2024\001\orbit-sp3\.

COD0MGXFIN_20240010000_01D_05M_ORB.SP3.gz

After decompression, its header identifies SP3-d, a 300-second sampling interval and the IGS20 reference-frame field.

9 File organization and observation postprocessing

9.1 Date folders and a flat folder

In Settings, Downloaded files offers By date (year / day of year) and All files in the save folder. The flat option omits date/category folders for ordinary files, while VMF retains model, grid and solution subfolders to separate identical filenames. Changing the layout does not move existing files. This setting is locked during active tasks.

Long filenames already contain dates. A flat folder can suit downstream tools that read files directly from one directory. When using several sources or expecting repeated names, test a small task first and check filename conflicts and the downstream application's directory requirements.

9.2 Extraction and Compact RINEX conversion

Enable Ask to extract and convert RINEX after downloading to receive one confirmation after the entire observation plan finishes. The option is off by default and does not apply to precise products. It processes files successfully downloaded or skipped as existing in the current task, rather than scanning and cleaning the whole save folder.

Supported archives include GZIP, Unix compress (.Z), ZIP, TAR / TAR.GZ, BZIP2, XZ and nested compression. The bundled CRX2RNX converts Compact RINEX .YYd, .d or .crx files to .YYo, .o or .rnx. A .crx.gz file, for example, is first extracted to .crx and then converted to .rnx. Extraction alone does not produce ordinary RINEX.

(1) Enable the postprocessing confirmation option before starting the observation task.

(2) When the plan finishes, read the confirmation dialog's file count, affected disk, recommended extra space and current free space.

(3) Confirm and wait for extraction, conversion and result commits. Source archives and intermediate Compact RINEX files are removed only after all corresponding results have been committed successfully.

(4) If space is insufficient, free space on the listed disk and click Retry. You can also cancel and retain the unprocessed originals.

(5) Inspect the final ordinary RINEX files. Check their headers, systems, observation codes and epoch ranges against the requirements of the next application.

The space recommendation is a conservative estimate that includes expanded data, conversion outputs, temporary work and a safety margin. Failure or cancellation retains originals that have not been processed successfully; other completed outputs remain available. Outputs with the same name but different contents are not overwritten. Investigate any conflict message before proceeding.

Figure 9.1 shows the actual confirmation for the single-file example. Space figures vary with the files and disk. After confirmation, ZIM2's .crx.gz was converted to the corresponding .rnx file with a RINEX 3.02 header. Postprocessing reported one success and no failures.

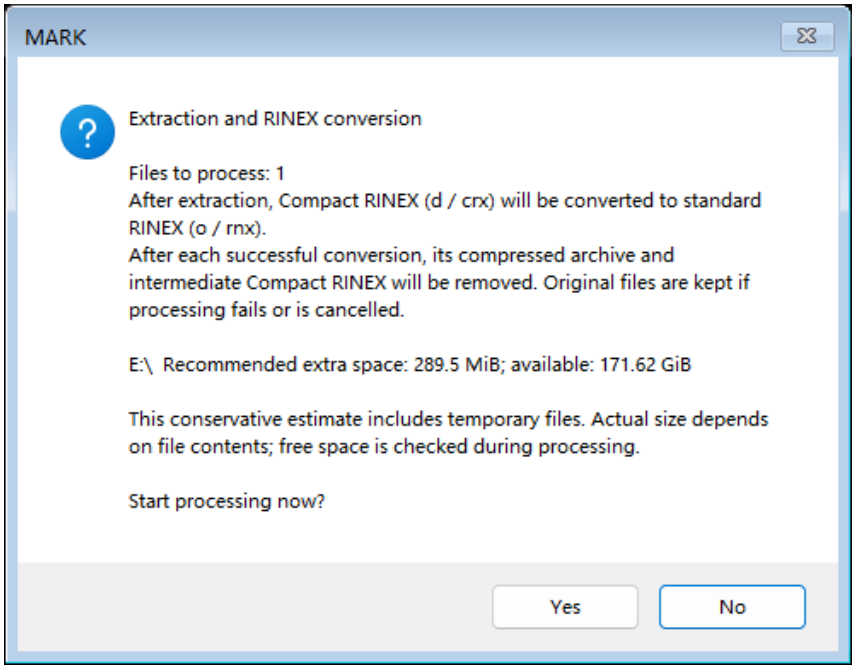


Figure 9.1 Confirmation for observation extraction and RINEX conversion

9.3 Validating results and recording experiments

First inspect task counts and confirm that files are nonempty. Check that archives can be decompressed and that file headers and time ranges are plausible. Check the RINEX version and file-type header; Compact RINEX

also contains CRINEX VERS / TYPE. Validate SP3, CLK, IONEX and Bias-SINEX products against their respective formats. Extensions and completion messages do not replace content checks.

Retain station or product IDs, dates, filters, the knowledge-base version, actual filenames, sources and SHA-256 hashes. For multi-file tasks, keep a file inventory and the station-coverage CSV to support reproducibility after knowledge-base updates or remote archive changes.

10 Complete worked examples

10.1 A full month with one receiver model and 60 stations

This example queries 2018-06-01 through 2018-06-30. Select all locally available sources, then apply interval 30S, receiver TRIMBLE NETR9 and 100% qualified date coverage. Select 60 stations with an even spatial spread. Record the counts returned by the installed knowledge base at each step.

Table 10.1 Successive filters and result checks

Step	Added condition	Result to check
1	Restore defaults; 30S; other restrictions disabled	Record the initial number of candidates.
2	Include receiver TRIMBLE NETR9	Record the remaining candidates with matching receiver evidence.
3	Qualified date coverage ratio ≥ 100%	Every effective date must pass file and receiver conditions.
4	Target 60; both distances 0; unrestricted daily availability; Even spread	Up to 60 stations, subject to coordinate and coverage constraints.
5	Check selected set	Inspect the confirmed daily counts and any failing dates.

(1) Clear the station search and spatial scope, and disable space-weather and quality filters. Select all available networks and set the full-month date range.

(2) Restore defaults in Advanced filter, confirm 30S, run the query and record the initial candidate count.

(3) Include receiver TRIMBLE NETR9, apply the filter and record the second count.

(4) Set qualified date coverage to Qualified date ratio ≥ with a threshold of 100. Apply and record the third count.

(5) Set Target count to 60, both distance limits to 0 and Daily availability to unrestricted, then click Even spread.

(6) Click Check selected set and inspect daily availability and spatial distribution. Export the report to retain the selected set, then add the required stations to a download plan.



Figure 10.1 An even-spread selection of 60 stations

The receiver filter can retain a station that qualifies on only some dates. Requiring 100% qualified date coverage then requires the file and receiver conditions on every effective date. Records without coordinates can appear in the table but not on the map. Read source-coverage and missing-coordinate notices when interpreting the counts.

10.2 Five product rules across 32 dates

Save a preset named Product filter example and set dates from 2024-01-01 through 2024-02-01, inclusive: 32 UTC dates. Table 10.2 and Figure 10.2 show its rules. This example demonstrates reusable filters for several product types. Select the actual series according to the research requirements.

Table 10.2 Custom preset rules

Type	Level	Center	Interval
SP3	Final	CODE	5 min
CLK	Final	GFZ	30 s
DCB	Rapid	CAS	1 d
OSB	Final	CODE	1 d
GIM	Final	IGS	2 h

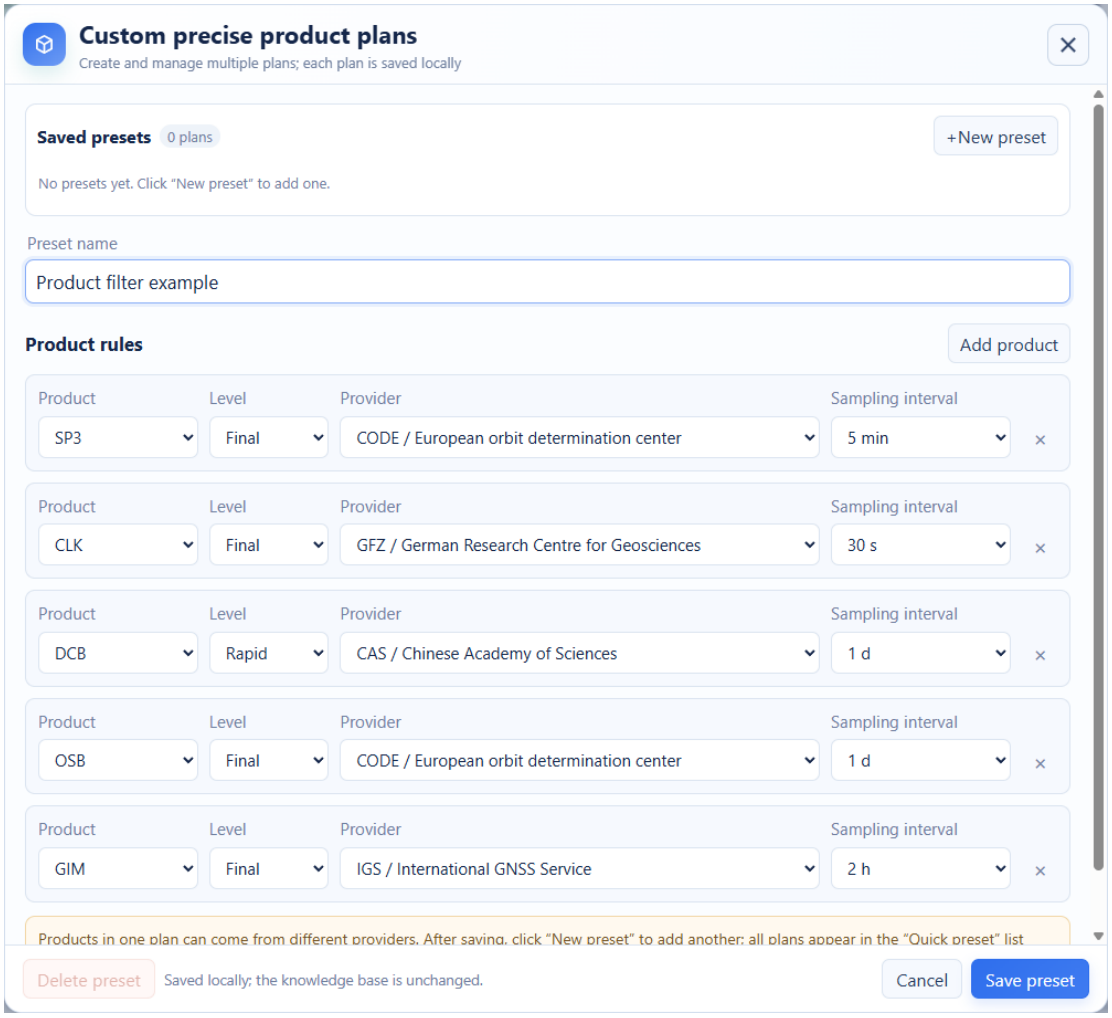


Figure 10.2 A custom preset with five product rules

Save the preset, select it and click Apply & query. A rule can match more than one product series. Inspect each row's ID, systems and date coverage. CODE OPS and MGX have different system coverage even where center and sampling interval match. Figure 10.3 illustrates the combined results from the five rules.

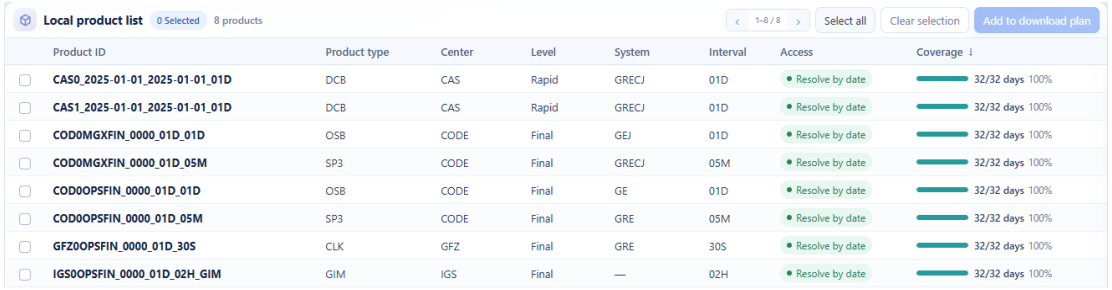


Figure 10.3 Product series matched by five rules

For a particular product line, use the IDs, systems and details to select the required rows before adding them to a plan. The example's joint display of CODE orbits and GFZ clocks does not establish a PPP bundle whose products have been verified as compatible.

10.3 Downloading one daily orbit file

(1) In Precise Products, set both dates to 2024-01-01, Product type to SP3, Product level to Final and Analysis center to CODE. Click Query products.

(2) Select COD0MGXFIN_0000_01D_05M from the CODE results and verify systems GRECJ and interval 05M.

(3) Click Add to download plan, confirm the dates and selected series, choose a save folder and start downloading.

(4) Inspect the completion counts and the original .SP3.gz file in 2024\001\orbit-sp3.

Use this product task and the observation task in Section 4.3 to check local searches, plan creation, file transfers and output paths. Transfer time depends on file size, network conditions and the provider. Inspect the task log and actual output for each run.

11 Troubleshooting

11.1 Startup and knowledge-base problems

Table 11.1 Troubleshooting startup and queries

Symptom	Checks and action
Application will not open or shows a blank page	Check Windows x64, WebView2 Runtime and the complete resources folder. Reinstall the complete package if required.
All sources are unavailable	Keep software and knowledge-base adjacent with all database subfolders. Use the official client distributed with the knowledge bases.
Query returns no results	Start with one available source and one indexed date. Clear search, spatial and advanced filters, then restore conditions individually.
More table records than map points	Inspect missing-coordinate notices. Records without coordinates cannot be plotted or used for distance-based optimization.
Equipment or system fields show a dash	Metadata is missing or not loaded. Do not infer models or capabilities; strict include filters require supporting records.
Incomplete source or unindexed-date notice	Check date coverage. Unknown records are not confirmed missing observations. Obtain an updated knowledge base from the maintainer if needed.

The release data cutoff is 2026-08-01 inclusive. A date within that limit can still lack records for a particular source, product or field. A query outside the supplied coverage is not evidence that files exist on the provider’s server. Check the indexed range before preparing a research dataset.

11.2 Transfer and postprocessing problems

Table 11.2 Troubleshooting downloads and files

Symptom	Checks and action
HTTP 404 or FTP 550	Record the station or product ID, UTC date, filename and source. Check whether the provider moved the archive or withdrew the file.
DNS, TLS or timeout errors	Check the system network, organizational proxy and reachable sources. MARK can use system static proxies and standard proxy environment variables.
Downloads stop or disk space is insufficient	Check folder permissions and free space. Postprocessing also needs space for expanded files and temporary work.
Existing file is skipped but cannot be read	MARK performs basic file checks, but these do not establish scientific completeness. Preserve and move aside a confirmed corrupt file, then download again.
No postprocessing prompt	Enable the option before the task and confirm that the task contains observations. Precise products do not trigger this step.
Output remains .crx or .YYd	It is Compact RINEX. Extraction must be followed by CRX2RNX conversion; inspect postprocessing results.
Conflicting output filename	Keep originals and compare contents, then process in a separate folder if needed. Different existing outputs are not overwritten.
SFTP or OpenSSH notice for a source	Follow the notice to check the Windows OpenSSH client and source access requirements. Retain the complete error.

MARK has no dedicated setting tied to a particular proxy node. Follow the provider's requirements for account-restricted or authorization-dependent sources. Successful access to another public source does not establish access to a restricted one.

11.3 Information to include in a problem report

Record the release, operating system, knowledge-base version, workbench, date range, station or product IDs, enabled filters, folder layout, success and failure counts, and complete Runtime errors. Screenshots should include relevant controls and status, rather than the word Failed alone. Do not publish passwords, access tokens or private credentials when sharing network settings.

12 Data updates and usage notes

12.1 Knowledge-base coverage and updates

The knowledge bases form a local release snapshot with data through 2026-08-01 inclusive. Individual sources and fields can end earlier or have gaps. Starting MARK does not automatically extend the snapshot. Install a compatible knowledge-base update supplied by the maintainer, retaining the expected folder structure and matching client.

The application source and knowledge bases have separate licenses. Read LICENSE and KNOWLEDGE_BASE_LICENSE.md before redistributing either component. Use compatible knowledge-base updates with the matching release client.

12.2 Research records and software citation

In a paper or report, record the MARK version and knowledge-base snapshot, query dates, sources, filters and selected stations or products. Cite GNSS data as required by the original providers. For space-weather filters, record index sources, daily statistics and thresholds. For GIM contributor exclusions, record providers, grades and the use of a station union across the date range.

Software: MARK v1.0.0

Knowledge-base data cutoff: 2026-08-01 inclusive

Query range: UTC start date through end date

Filters: sources, interval, equipment, quality, coverage and spatial conditions

Selection: station identifiers or product IDs

Results: task logs, file inventory, sources, access dates and SHA-256 hashes

Once the software paper is published, use the bibliographic information provided on the project release page. Data-provider and analysis-center citation requirements still apply separately. Citing MARK does not replace attribution for the data.

12.3 Building from source

Developers need Visual Studio 2022 with the Desktop development with C++ workload and Windows SDK, plus CMake 3.24 or later. See docs/BUILD.md in the source repository for the build procedure and full requirements.

Build outputs are placed in build-private\Release\. Ordinary users can run the official release executable without building the application.

Appendix A Terms and filename conventions

Table A.1 Common terms

Term	Meaning
GNSS	Global Navigation Satellite System.
IGS / MGEX	International GNSS Service and its multi-GNSS data view.
UTC / DOY	Coordinated Universal Time / day of year.
RINEX	Receiver Independent Exchange Format; observation and navigation files have distinct roles.
Compact RINEX	Hatanaka-compressed observations, commonly .crx or .YYd, requiring dedicated conversion.
Site Log	Station and equipment records; these do not replace file availability for a specific date.
Product series	A logical entry for a product class that can be resolved to files by date.
Analysis center / download source	Product producer / institution or archive route supplying file transfers.
Completeness / unverified	Date coverage within the requested range / a record whose availability is not confirmed.

Term	Meaning
GRECJ	Common system letters: G = GPS, R = GLONASS, E = Galileo, C = BeiDou, J = QZSS.

In the observation filename ZIM200CHE_R_20240010000_01D_30S_MO.crx.gz, ZIM200CHE is the long station identifier. The time field 20240010000 encodes the start of DOY 001 in 2024. The fields 01D and 30S indicate duration and sampling interval, MO denotes mixed-system observations, and crx and gz indicate Compact RINEX and GZIP compression. Consult the relevant RINEX specification and actual header for complete definitions.

In COD0MGXFIN_20240010000_01D_05M_ORB.SP3.gz, COD is a CODE-related center code, MGX identifies the product line and FIN the final level. The time field gives the start time, 01D and 05M the duration and interval, and ORB.SP3 the orbit product. Filename codes and interface names do not always map one to one; verify the details before downloading.

Appendix B Common data-preparation tasks

Table B.1 Tasks and recommended operations

Task	Recommended operation
Short observation task for one station	Select sources and UTC dates, set format and interval, search the station ID and add a plan.
Full-month equipment filtering	Combine interval, equipment include filters and 100% qualified date coverage, then inspect actual files.
Global evaluation station set	Widen sources, verify contributor-exclusion rules, use Even spread and inspect daily spatial coverage.
Regional candidate network	Set countries or coordinate bounds, distance and daily constraints. Check graph connectivity and available coverage.
Standard PPP data preparation	Prepare observations, then query a built-in PPP preset for the same dates and check the processing requirements.
Repeated multi-product queries	Save a custom preset and apply it after each date change. Select series and systems explicitly.
Application requiring ordinary RINEX	Enable observation postprocessing and verify extraction and Compact RINEX conversion outputs.